Project Report

Hindi Playfair Cipher

A Specialized Cryptographic Tool for Devanagari Script

1. Introduction

In the contemporary digital landscape, cryptography is a cornerstone of information security. While traditional methods often focus on Latin-based scripts, there is a growing need for security tools that accommodate linguistically diverse systems. This project presents a specialized adaptation of the **Playfair Cipher**, a symmetric encryption technique that processes pairs of characters (digraphs).

The **Hindi Playfair Cipher** is specifically engineered to address the phonetic and structural complexities of the Devanagari script, providing a robust framework for securing Hindi text.

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2. System Architecture

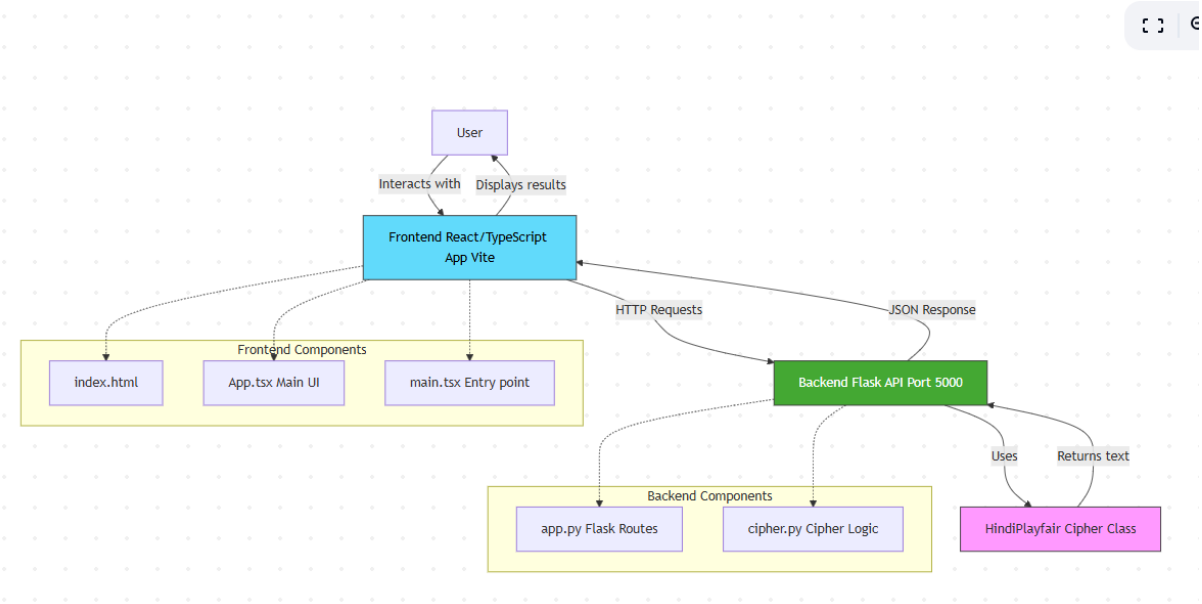
The application utilizes a sophisticated full-stack architecture designed for scalability, performance, and a clear separation of concerns between the user interface and the core cryptographic logic.

- **Frontend Layer**: Developed using **React, Vite, and TypeScript**, the frontend offers a high-performance environment. It features a real-time,

animated interface that allows users to visualize the encryption and decryption processes instantaneously.

- **Backend Layer**: A **Python Flask** server serves as the engine of the application. It manages all cryptographic operations and exposes them to the frontend via a secure RESTful API.

- **UI/UX Design**: To ensure a premium and responsive experience across all devices, the interface is styled with **TailwindCSS** and enhanced with **Framer Motion** for fluid transitions.



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3. The 7x7 Hindi Grid Algorithm

A significant innovation of this project is the expansion of the traditional 5×5 English Playfair grid into a **7x7 matrix**. This modification provides 49 distinct slots, which is essential for capturing the expansive character set of the Devanagari script:

Category	Characters Included
Vowels (11)	अ, आ, इ, ई, उ, ऊ, ऋ, ए, ऐ, ओ, औ

Category	Characters Included
Consonants (33)	क through ह
Special Characters (5)	अं, अः, ् (Halant), क्ष, ज्ञ

Padding Logic: The character **'क्ष'** serves as the designated filler, used for padding odd-length strings and handling double-letter occurrences within a digraph.

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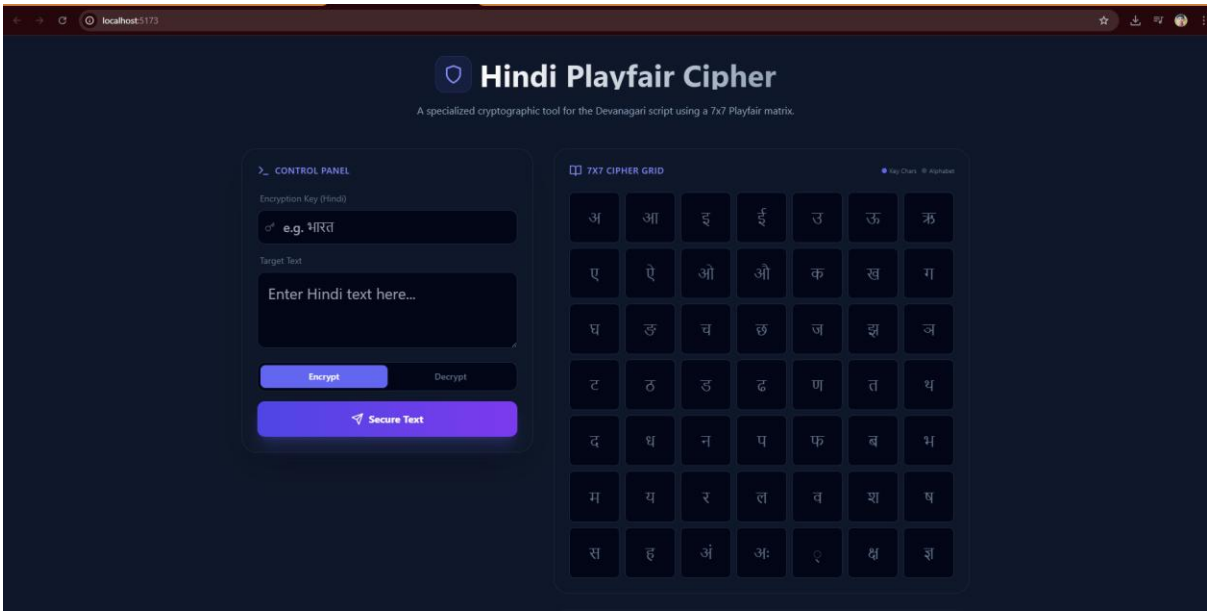
4. Operational Logic

The system adheres to three primary geometric rules for character transformation within the 7x7 grid:

1. **Same Row Rule**: Characters are replaced by the neighbor to their immediate right, with horizontal wrap-around.

2. **Same Column Rule**: Characters are replaced by the neighbor immediately below, with vertical wrap-around.

3. **Rectangle Rule**: For characters forming a rectangle, each is swapped with the corner character occupying the same row within that rectangle.



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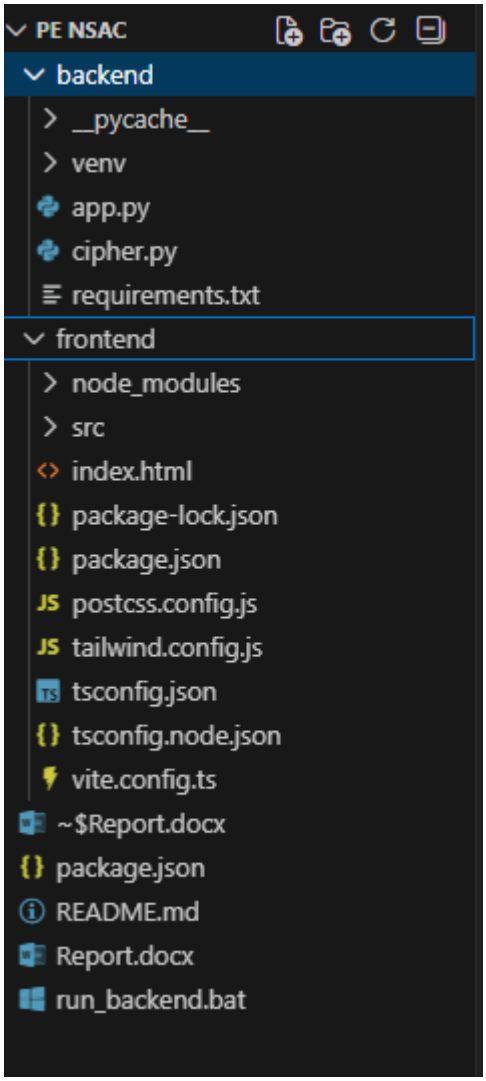
5. Technical Stack Summary

The following table outlines the specialized technologies integrated into this solution:

Category	Technology Used
Frontend Framework	React (Vite) + TypeScript
Styling & Animation	TailwindCSS + Framer Motion
Backend Environment	Python (Flask)
Cryptographic Core	Custom 7x7 Playfair Algorithm



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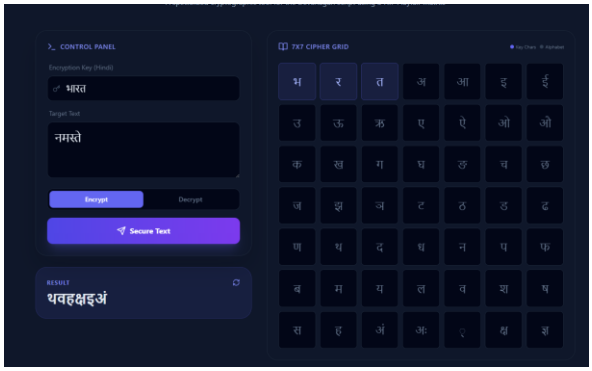
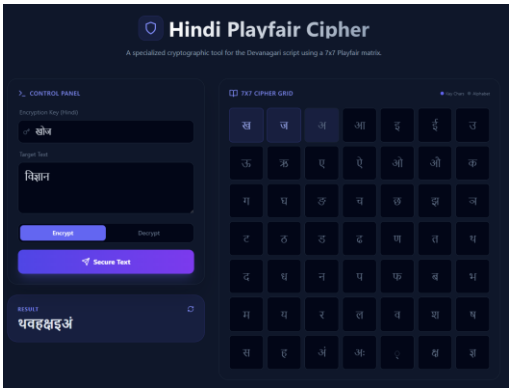
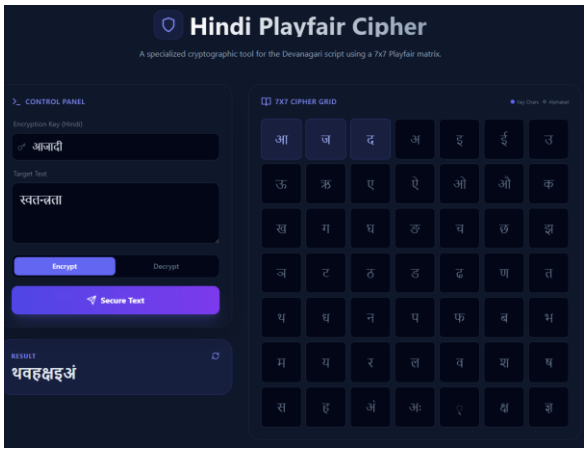
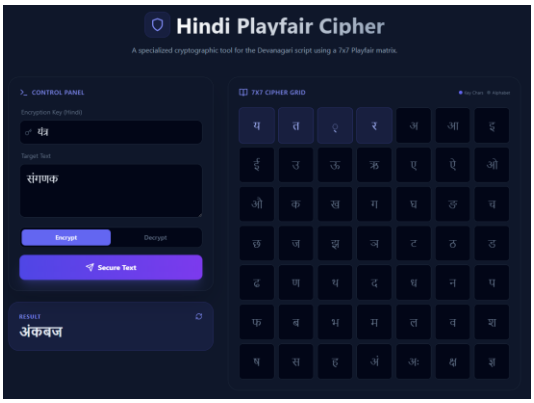
6. EXAMPLES:-

The **Hindi Playfair Cipher** successfully bridges the gap between classic cryptographic principles and linguistically complex scripts.









6. Conclusion

The **Hindi Playfair Cipher** successfully bridges the gap between classic cryptographic principles and linguistically complex scripts. By leveraging a modern, modular tech stack, the project provides a secure, user-friendly, and highly efficient method for processing Devanagari text, demonstrating that traditional ciphers can be effectively modernized for global languages.